Challenges in Sustainability
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Developing Probabilistic Scenarios for Strengthening Tourism Village Resilience Under Sustainability Challenges in the Mount Merapi Tourism Area, Indonesia

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Received: 05-15-2026 **Revised:** 09-17-2026 **Accepted:** 09-24-2026

Citation: Juwita, A. H., Fauzi, A., Rustiadi, E., & Rindayati, W. (2026). Developing probabilistic scenarios for strengthening tourism village resilience under sustainability challenges in the Mount Merapi tourism area, Indonesia. *Chall. Sustain.*, *14*(5), 1019–1037. https://doi.org/10.56578/cis140509.

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Abstract: Tourism villages surrounding Mount Merapi have demonstrated remarkable resilience, recovering from various disruptions and periods of inactivity. However, these communities continue to face significant sustainability challenges that threaten their long-term viability. These challenges stem from both external factors—particularly their location in a disaster-prone volcanic region—and internal factors related to local community capacity and tourism management capabilities. Building and maintaining resilience in these tourism villages is therefore crucial for ensuring sustainable tourism development in the region. This research develops probabilistic scenarios to assess tourism village resilience in the Mount Merapi area of Yogyakarta Special Region, Indonesia, employing a participatory Bayesian Network approach. The study classified administrative villages into two infrastructure-based clusters and selected ten representative tourism villages across both clusters. While the same Directed Acyclic Graph (DAG) framework was applied to both clusters, probability assessments were conducted independently through cluster-specific focus group discussions (FGDs). Key findings reveal that Cluster B demonstrates a higher baseline probability of achieving "Stable" resilience compared to Cluster A. Through strength analysis, government involvement and internal policy frameworks emerged as critical factors with significant interconnections to other variables. Sensitivity analysis identified economic activities, agrotourism development, and cultural preservation as particularly influential variables in both clusters, though their probability conditions differ between clusters. Forward propagation analysis demonstrated how specific states of sensitive variables affect the likelihood of achieving "Stable" resilience in both clusters. Conversely, backward propagation analysis identified the probability conditions associated with the desired "Stable" resilience state. The study concludes that enhancing tourism village resilience requires an integrated approach that simultaneously addresses economic activities, agrotourism development, cultural preservation, social capital, internal policy frameworks, government roles, and natural disaster response. This holistic strategy is essential for these communities to successfully navigate future sustainability challenges and maintain their resilience in an inherently volatile environment.

Keywords: Resilience; Tourism village; Rural tourism; Sustainability; Community engagement

1. Introduction

Resilience refers to a system's capacity to confront and adapt to dynamic challenges from external or internal sources while maintaining its core functions through innovative approaches or resource reorganization (Espiner et al., 2017; Wu et al., 2025). Contemporary research across economic, social, cultural, and environmental domains increasingly recognizes resilience as essential for achieving sustainable long-term development. The tourism

sector particularly requires resilience due to its high susceptibility to disruptions including natural disasters, economic downturns, climate change, and pandemics that can jeopardize destination sustainability. This study adopts an evolutionary perspective on tourism village resilience, emphasizing not merely recovery to pre-shock conditions but the capacity to adapt, reorganize resources, and sustain tourism functions amid changing circumstances (Gaki & Koufodontis, 2022; Yu et al., 2023).

The transition from conventional to rural tourism gains broader recognition as a sustainable tourism model. Rural tourism development integrates tourism sustainability, community-based approaches, cultural and environmental conservation, and local economic strengthening (Ruiz-Ballesteros & González-Portillo, 2024). Research indicates that sustainable tourism should bolster community resilience rather than depend on it (Espiner et al., 2017). Consequently, rural tourism—functioning as both sustainable tourism and local economic activity—becomes central to community resilience by providing alternative income sources during external shocks (Handmer & Choong, 2006). Since local communities directly manage these activities, rural tourism serves strategically in supporting economic development while building community capacity to respond and adapt to various disruptions.

Research identifies several critical factors for tourism destinations to survive and continue operating post-shock. Maintaining consistent revenue flows from visitors or external support proves essential for sustaining tourism activities and building recovery capacity (Espiner et al., 2017). Studies across South Africa, China, and neighboring regions emphasize strengthening institutions and stakeholder-tourist governance during crises (Kupika & Dube, 2023; Wang et al., 2022). Research by Saskara et al. (2025) highlights the importance of traditional institutions and government support for weaving industry sustainability, while Tian et al. (2024) and Wang et al. (2022) note the ongoing need for governance reviews in sustainable tourism resilience. Additional recommendations include establishing financial reserves (Kupika & Dube, 2023) and maintaining tourist villages through adaptive reuse and circular economy principles without compromising local culture and environment (Makwindi & Ndlovu, 2022).

These investigations reveal that tourism resilience encompasses multiple dimensions: financial capacity, governance, institutional support, cultural and environmental preservation, and adaptive strategies (Espiner et al., 2017; Kupika & Dube, 2023; Makwindi & Ndlovu, 2022; Tian et al., 2024). However, previous studies focus on different factors and varying shock or recovery conditions. This creates a gap in understanding how these factors interact and how their combined influence affect community-managed rural tourism resilience under uncertainty—particularly crucial for tourism villages in disaster-prone areas where resilience involves both recovery from past shocks and preparation for future disruptions.

In Indonesia, rural tourism manifests as tourism villages that leverage local community capabilities to manage resources while preserving environmental and cultural integrity (Ariyani et al., 2023). Tourism villages on Mount Merapi's slopes face distinctive sustainability and resilience challenges. Their appeal stems from natural resources and community traditions, with attractions including river walks, rice fields, plantations, former eruption sites, volcanic tourism, traditional performances, local customs, and village life experiences.

Over two decades, Mount Merapi tourism villages have encountered major external shocks. The 2010 eruption achieved Volcanic Explosivity Index (VEI) IV, producing approximately 130 million m^3 of volcanic material and severely disrupting communities and tourism (Ministry of Energy & Mineral Resources of Republic Indonesia, 2020). Previous research documented how local tourism communities adapted to eruption impacts. The COVID-19 pandemic presented another significant shock. Sleman Regency Tourism Office data shows northern area tourist visits (including Mount Merapi tourism zones) grew 891.23% in 2019 but plummeted -79.80% in 2020. While recovery occurred with 30.49% growth in 2024 and 7.75% in 2025, patterns indicate uneven tourism recovery compared to pre-pandemic levels (Sleman Regency Tourism Office, 2026).

Mount Merapi tourism villages continue facing ongoing challenges. Volcanic activity remains an uncertainty source, while disaster preparedness and risk communication remain vital for tourism management and visitor safety (Rindrasih, 2018; Wijaya et al., 2025). Mount Merapi's volcanic impact on tourism varies temporally major eruptions can halt tourism requiring extended recovery periods, while lower activity levels permit continued tourism with hazard-zone restrictions. These conditions make Mount Merapi an important resilience case study, having experienced major shocks and recovery while confronting persistent future hazards. This enables examining resilience not only as post-shock recovery but as the capacity to maintain tourism activities, prepare for future disturbances, and adapt under continuous uncertainty—practically important for sustaining local livelihoods, tourist safety, and tourism activity continuity (Nurjanah et al., 2026; Rindrasih, 2018).

Although resilience is dynamic and multidimensional, this study operationalizes tourism-village resilience as two end states—stable and unstable—to support Bayesian Network analysis. A stable state refers to conditions in which tourism-village activities can continue or recover after a disruption while restoring core functions. An unstable state refers to conditions in which tourism activities cease or become dormant and struggle to regain core functions. Accordingly, "Stable" and "Unstable" are treated as model target states rather than a comprehensive definition of tourism resilience.

Tourism-village resilience is shaped by multiple interrelated factors and considerable uncertainty, requiring an

approach that can capture dependencies and probabilistic effects. A Bayesian Network is appropriate because it represents relationships among connected variables using probabilities, is well suited to complex and uncertain systems (Aguilera et al., 2011; Phan et al., 2016) and can integrate empirical data with expert or stakeholder knowledge through participatory model development (Castelletti & Soncini-Sessa, 2007; Phan et al., 2016). This aligns with the context of tourism villages, where resilience depends on interacting factors as well as the experience of managers and local stakeholders. Bayesian Networks also support scenario testing by examining how changes in conditions affect the likelihood of resilience outcomes (Singto et al., 2020). For these reasons, this study applies a participatory Bayesian Network to develop resilience stability scenarios for tourism villages in the Mount Merapi area.

This study contributes by developing probabilistic resilience stability scenarios for tourism villages in the Mount Merapi tourism area using two complementary analyses. The first scenario applies forward propagation by fixing selected variable nodes based on stakeholder agreement and sensitivity analysis. The second applies backward propagation by setting the resilience target node to a desired state. Using these scenarios, the study evaluates how shifts in economic activity, cultural preservation, social capital, government roles, and natural disaster response influence the probability of resilience. The findings are intended to offer actionable scenario alternatives for managers and stakeholders to strengthen resilience stability and support the long-term sustainability of tourism villages.

2. Methodology

2.1 Analysis Method

This study applied a Bayesian Network, a graphical model based on a Directed Acyclic Graph (DAG) that represents relationships among variables, including qualitative components (Stepshenson, 2000). In decision support systems (DSS), Bayesian Networks are commonly used to support management under uncertainty and to adapt to changing conditions such as visitor preferences, loyalty, arrivals, and behavior (Ariyani & Fauzi, 2024). The DAG structure of the Bayesian Network is illustrated in Figure 1.

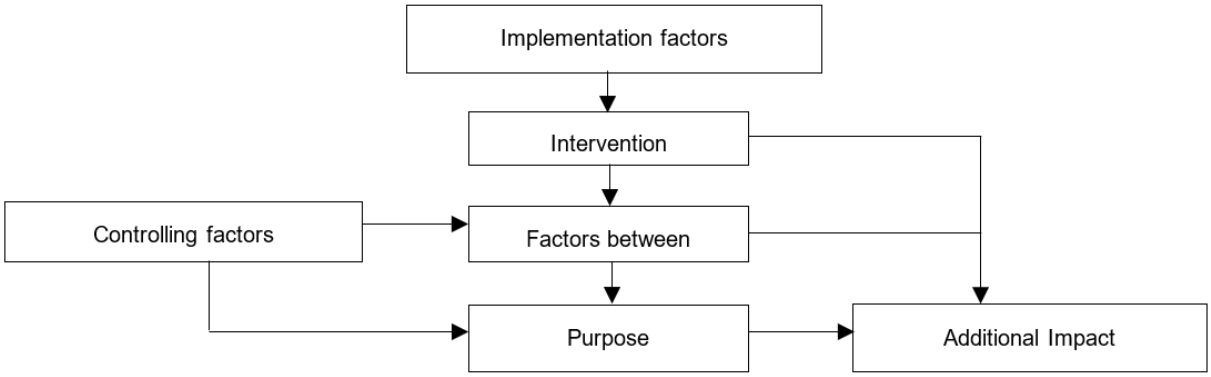


Figure 1. Structural Network
Source: Adapted from Cain (2001).

Based on the Figure 1, the purpose category refers to outcomes the researcher aims to influence through tourism-village resilience, either to improve conditions or prevent deterioration. These outcomes provide the basis for defining the resilience objectives. Intervention variables refer to actions or measures applied to achieve the objective, while Intermediate factors link interventions to objectives. Controlling factors cannot be directly altered through interventions but may influence the system in different ways. Implementation factors determine whether an intervention can be successfully implemented at present or in the future. Additional impacts refer to changes triggered by interventions that do not directly affect the system (Cain, 2001).

According to Tang et al. (2019), the Bayesian Network method is generally divided into three types of nodes in a particular Bayesian Network: (1) root nodes, which have no incoming edges or parent nodes; (2) leaf nodes, which have no outgoing edges or child nodes; and (3) Intermediate nodes, which have both incoming and outgoing edges. A basic Bayesian Network structure involving five variables is illustrated in Figure 2.

In Figure 2, the arrow goes from Z_1 to Z_2, showing that Z_1 is a parent's node of Z_2, while Z_2 is a child node of Z_1. Similarly, Z_2 is a parent node Z_3 and Z_4, while Z_3 and Z_4 are parent nodes of Z_5. Thus, the probability distribution of a child node is conditional on the states of its parent node or nodes. The systematically can be written:

$$P(Z_1, Z_2, Z_3, Z_4, Z_5) = P(Z_1)P(Z_2|Z_1)P(Z_3|Z_2)P(Z_4|Z_2)P(Z_5|Z_3, Z_4) \quad (1)$$

For network containing *n* variables, the full joint probability distribution can generally be expressed as:

$$P(Z_1, Z_2, \ldots . Z_n) = \prod_{i=1}^{n} P(Z_i \,|Pa(Z_i)) \quad (2)$$

where, $Pa(Z_i)$ represents the parent node or set of parent nodes of Z_i.

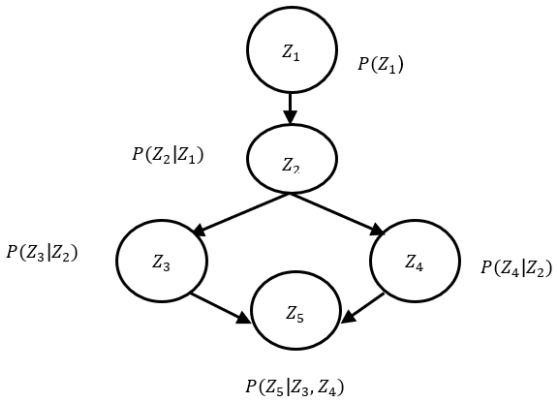


Figure 2. Bayesian Network illustration
Source: Adapted from Tang et al. (2019).

2.2 Data Analysis

2.2.1 Sampling and clustering process

Sampling used a cluster-based approach. First, the 30 tourism villages in the northern Mount Merapi tourism area were mapped to the administrative villages in which they are located—Bangunkerto, Umbulharjo, Mororejo, Pondokrejo, Candibinangun, Kepuharjo, Purwobinangun, Wonokerto, and Wukirsari. Consequently, the 30 tourism villages were not treated as clustering units. Clustering was conducted at the administrative-village level because infrastructure data were available and consistently reported at this scale.

Administrative villages were grouped using infrastructure indicators relevant to tourism village activities: household electricity availability, number of hotels, number of accommodation facilities (lodgings), number of cellular networks, availability of savings and loan cooperatives, number of shops, and number of restaurants. The data were drawn from the 2025 district-level publications (Statistics of Sleman Regency, 2025). Indicator selection was constrained by the availability of secondary data that were consistently reported across the study area. Variables such as tourism development level, disaster exposure, accessibility, and visitor numbers were not included because comparable data were not available for all administrative villages.

Clustering was performed using K-means in Orange Data Mining. Prior to analysis, variables were standardized using the Normalize Features function ($\mu = 0$, $\sigma^2 = 1$) to minimize the influence of differing measurement scales. The number of clusters was set to two to enable comparison between tourism villages under two broad infrastructure conditions: relatively limited versus relatively adequate supporting infrastructure. Cluster characteristics were interpreted from the standardized indicator profiles. The initial K-means output produced Cluster 1 (C1) and Cluster 2 (C2); for presentation, these were relabeled so that C2 became Cluster A and C1 became Cluster B. Cluster A represented administrative villages with infrastructure that was relatively less supportive of tourism village activities, while Cluster B represented administrative villages with relatively better supporting infrastructure. The clustering results and Silhouette values were then used as one consideration in selecting tourism villages as research samples (Table 1).

Based on Table 1, Cluster A comprises four administrative villages, while Cluster B consists of eight administrative villages. Silhouette values were used to assess how well each administrative village fit into its assigned cluster. A Silhouette value closer to 1 indicates a stronger fit within the cluster (Ikotun et al., 2023; Khan et al., 2024). For example, Mororejo had the highest Silhouette value in Cluster A, while Purwobinangun had the highest value in Cluster B. The clustering results and Silhouette values were then used as considerations in selecting tourism villages as research samples. The clustering process was first conducted at the administrative-village level, and tourism villages located within these administrative villages were then considered for sample selection. However, there are 30 tourism villages in the study area, and the number of tourism villages in each administrative village is different. Therefore, the final selection of tourism villages was not based only on the

Silhouette values. Consultation with stakeholders from the Tourism Office was also conducted during the sample-selection stage to confirm the suitability of the selected tourism villages based on their actual conditions in the field. A total of 10 tourism villages were selected to provide cases from both clusters while maintaining a feasible scope for field observation, interviews, and stakeholder engagement.

Table 1. K-means result

Cluster A	Silhouette	Standardized Infrastructure						
		Elect	Ho	Lo	CelNet	Loan	Shops	Rest
Kepuharjo	0.564	-1.343	-0.447	-0.461	0.361	-0.297	1.353	0.957
Candibinangun	0.635	-0.712	-0.447	0.079	-1.806	-0.042	2.435	-0.638
Pondokrejo	0.636	-1.023	-0.447	-0.461	-1.083	-0.552	0.271	-0.957
Mororejo	0.647	-1.388	-0.447	0.393	-1.806	-0.424	0.271	-0.957
Cluster B								
Merdikorejo	0.470	-0.830	-0.447	-0.461	0.361	-0.297	-0.812	-0.319
Wukirsari	0.527	0.159	-0.447	-0.124	1.083	-0.424	0.271	2.232
Bangunkerto	0.532	0.749	-0.447	-0.461	0.361	3.267	-0.812	-0.957
Girikerto	0.545	0.095	-0.447	-0.461	0.361	-0.424	-0.812	0.638
Umbulharjo	0.549	1.585	2.236	3.248	-0.361	-0.170	-0.812	1.276
Wonokerto	0.562	0.894	-0.447	-0.461	0.361	0.085	-0.812	-0.638
Donokerto	0.563	0.423	2.236	-0.056	1.083	-0.297	-0.812	-0.638
Purwobinangun	0.564	1.389	-0.447	0.011	1.083	-0.424	0.271	0.000

Note: elect, availability of electricity for households; Ho, number of hotels; Lo, number of accommodation facilities-lodgings; CelNet, number of cellular networks; loan, availability of savings and loan cooperatives; shops, number of shops; rest, number of restaurants.

Based on this process, 10 tourism villages were selected as research samples representing the characteristics of the two infrastructure clusters. In Cluster A, the samples consisted of Plumbon Lor Tourism Village in Mororejo, Lembah Krasak Jlapan in Pondokrejo, Sambirembe Tourism Village in Candibinangun, and Batur Berdikari in Kepuharjo. Meanwhile, in Cluster B, the samples consisted of Turgo Merapi Tourism Village in Purwobinangun, Pentingsari Tourism Village in Umbulharjo, Sangurejo in Wonokerto, Agrowisata Turi and Plosokuning in Bangunkerto, and Bulaksalak in Wukirsari.

2.2.2 The focus group discussion procedure

Based on Figure 3, the steps taken before focus group discussion (FGD) procedure are as follows:

(1) Variable identification and developing Directed Acyclic Graph

The initial variables and proposed relationship between variables were identified from observation and previous literature related to tourism resilience, including economic activities, social capital, institutional and government roles, tourism management, and natural disaster response (Ariyani & Fauzi, 2024; Cain, 2001; Farsari, 2023; Gocer et al., 2024; Rindrasih, 2018; Wijaya et al., 2025). The conceptual basis underlying in the relationships among the variables are presented in Table 2. The variables were grouped based on their role in the Bayesian Network structure, consisting of objectives, interventions, intermediate factors, controlling factors, implementation factors, and additional impact variables, as presented in Table 3.

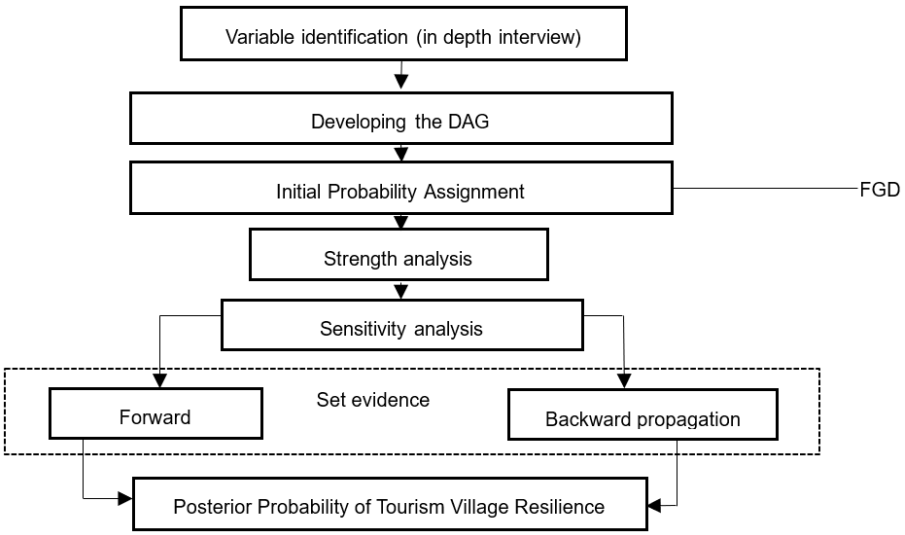


Figure 3. Research flow using Bayesian Network

Table 2. Relationship in Directed Acyclic Graph (DAG)

Relation of Edge		Literature
Internal policies	Tourism awareness group Economic activities Agrotourism	Abdul Khoir & Dirgantara (2020) provide conceptual support for the relationships between internal policies, tourism awareness groups, economic activities, and agrotourism. Their discussion highlights the role of regulations and community-based tourism management in supporting tourism development, local economic empowerment, tourism awareness, and the use of local environmental potential for tourism activities.
Social capital	Cultural preservation Tourism awareness group Agrotourism	These relationships are conceptually supported by Gocer et al. (2024), particularly through their discussion of social capital and cultural heritage. In this study, these relationships were also supported by field observations and confirmed through interviews. Gocer et al. (2024) describe social capital in terms of the social processes and activities that enable people and groups to come together in a particular place. Collective activities and projects involving community members can strengthen the relationships among them. This relationship is conceptually supported by Farsari (2023), who explains that the traditional role of public policy and government is to encourage and regulate tourism development through subsidies or taxes, marketing, and planning.
The role of government	Natural disaster response Cultural preservation Tourism awareness group Agrotourism	This relationship is supported by Wijaya et al. (2025), who state that collaboration between organizations and government agencies is important for developing effective strategies and improving performance during disasters. They also report that tourism managers consider preparedness for Mount Merapi eruptions to require coordination and collaboration with various parties, including village governments, disaster preparedness groups, the Centre for Geological Disaster Technology Research and Development (BPPTKG), the Regional Disaster Management Agency, and the Tourism Office. This relationship is indirectly supported by Ariyani & Fauzi (2024) who discuss the role of government support and rural tourism development. They note that government, through public funding and other public-sector partners, can help support key operations in the accommodation sector during times of crisis, while rural tourism contributes to economic development and poverty alleviation by offering authentic rural experiences, highlighting natural beauty, and promoting cultural heritage. This relationship is indirectly supported by Ariyani & Fauzi (2024). Their study notes that government and other public partners can support important tourism operations during crises. They also explain that rural tourism offers authentic rural experiences, highlights natural beauty, and promotes cultural heritage.
Agrotourism	Resilience of tourism village Economic activities	This relationship is conceptually supported by Ariyani & Fauzi (2024), who describe how community groups developed tourism attractions inside, outside, and around the reservoir area as public awareness of tourism potential increased. They also explain that these developments were related to changes in local livelihoods, particularly among farmers whose agricultural land was affected by the construction of the reservoir. This relationship is supported by the concept discussed by Ariyani & Fauzi (2024), who state that rural tourism represents a form of productive diversification that generates additional economic income for farmers and may also encourage the conservation of uncultivated semi-natural environments.
Economic activities	Resilience of tourism village	This relationship is supported by Gocer et al. (2024), who explain that rural economic diversification and growth through new employment opportunities, tourism businesses, and entrepreneurship provide community-level economic opportunities that can reduce rural vulnerability and enhance community resilience.
Natural disaster response	Cultural preservation	Wijaya et al. (2025) state that Merapi tourism managers consider disaster preparedness to be strongly influenced by support from other partners involved in disaster preparedness and planning. Furthermore, Rindrasih (2018) discusses disaster response and recovery in the context of Community-Based Tourism (CBT), emphasizing that CBT depends on the local natural environment and culture and that local cultural identity should be preserved throughout tourism activities. Therefore, these concepts are used as indirect support for the relationship between natural disaster response and cultural preservation.
Cultural preservation	Resilience of tourism village	This relationship is conceptually supported by Ariyani & Fauzi (2024), who explain that rural tourism offers authentic rural experiences, highlights natural beauty, and promotes cultural heritage. It is further supported by Gocer et al. (2024), who discuss the importance of resilience in the context of sustainable development and heritage conservation.

Table 3. Bayesian Network structural variables

Variable Category	Variables Nodes	Conceptual Meaning	Measurement	Definition
Objectives	Resilience of tourism village	Resilience of tourism villages is the ability of tourism villages to maintain and recover their tourism activities when facing internal or external shock (Espiner et al., 2017).	Unstable	Tourism activities stop or enter a dormant period after a disruption and have difficulty resuming their main functions.
			Stable	Tourism activities can continue or resume after a disruption while restoring their main functions.
Interventions	The role of government	Government involvement in supporting the management and sustainability of tourism village (Farsari, 2023; Kupika & Dube, 2023).	No	Government support, coordination, training, or assistance is absent or very limited.
			Yes	Government provides support through coordination, training, assistance, policy, or tourism development programs.
	Internal policies	Internal rules and decisions used by tourism village managers to manage tourism activities, example AD/ART (Abdul Khoir & Dirgantara, 2020).	No	There are no clear internal rules, division of responsibilities, or management procedures.
			Yes	Internal rules, division of responsibilities, or management procedures are available and implemented.
Intermediate factor	Tourism awareness group	The involvement of tourism awareness groups in managing and developing tourism village activities (Abdul Khoir & Dirgantara, 2020; Ariyani & Fauzi, 2024).	Low	Participation and activities are limited and not regularly conducted.
			Moderate	The group participates in several tourism activities, but its involvement is not continuous.
			High	The group actively participates in tourism planning, management, promotion, and community activities.
	Social capital	Social relationships, trust, cooperation, and community participation that support tourism village management (Gocer et al., 2024).	Developed	Cooperation and social networks have developed but still depend on actors or activities. Tourism village activities still rely on initiatives driven by these actors.
			Built in	Cooperation, trust, and collective participation are embedded in community activities and regularly support tourism management. Tourism village activities can continue through established community networks without depending on specific actors.
Controlling factors	Natural disaster response	The ability of tourism village managers and communities to prepare for and respond to natural disaster risks (Rindrasih, 2018; Wijaya et al., 2025).	Unprepared	Disaster preparedness is limited, such as limited evacuation information, training, early warning information, or community response mechanisms.
			Preparedness	Evacuation routes or procedures, disaster information, training, early warning mechanisms, or community response mechanisms are available and understood.
Implementation factors	Agrotourism	The use of agricultural resources and activities as part of tourism products or visitor activities (Ariyani & Fauzi, 2024).	Bad	These resources are available but have not been adequately developed or integrated into tourism activities. There are no tourism activities that use agrotourism resources.
			Moderate	These activities are included in tourism offerings, but their development remains limited. The activities mainly introduce traditional farming practices, have limited variety, and are often dependent on favorable weather conditions.

			Good	These activities are actively managed and integrated into tourism products or visitor experiences. Tourist not only introduced to traditional farming practices but also provided with opportunities to participate and processing products.
	Cultural preservation	Community efforts to maintain local traditions, arts, and cultural activities as part of tourism village resources (Ariyani & Fauzi, 2024).	Passive	Cultural traditions still exist but are rarely practiced, promoted, or involved in tourism activities.
			Active	Cultural traditions are regularly practiced, preserved, promoted, or included in tourism activities.
Additional impact variable	Economic activities	Economic activities generated or supported by tourism and involving local communities (Espiner et al., 2017; Gocer et al., 2024).	Limited	Tourism-related economic activities are limited to one sector.
			Adequate	Several tourism-related economic activities are available across multiple sectors, including accommodation, food services, MSMEs, local products, tourism services, or local employment.

Based on these variables and their relationships, the initial DAG was developed from observation and previous literature. In-depth interviews were then conducted with tourism village managers and representatives from the Tourism Office to confirm the relevance of the selected variables and the relationships between variables based on the actual conditions of tourism villages in the Mount Merapi area. During the in-depth interviews, no variables or relationships in the initial DAG were rejected because the participants considered the proposed structure relevant to the actual conditions of tourism villages. The documentation regarding the differing views of the stakeholders on this stage was not recorded separately.

The confirmed DAG structure is presented in Figure 4. The same DAG structure was used for both clusters. After this stage, the DAG was presented during the FGD, and the participants assessed the probability of each variable relationship based on the measurement conditions presented in Table 3.

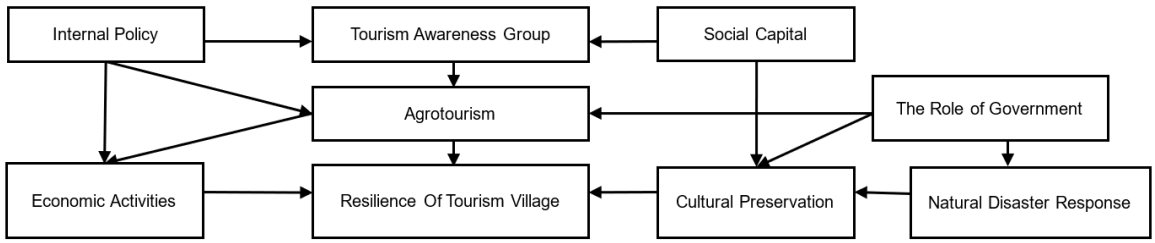


Figure 4. Directed Acyclic Graph (DAG) structure Bayesian Network resilience tourism village

(2) The initial probability assessments through focus group discussion

The FGD involved 20 participants who had knowledge of tourism village management and development. The participants consisted of two representatives from the Sleman Regency Tourism Office, two representatives from the Sleman Regency Development Planning Agency, four members of tourism awareness groups, ten tourism village managers, and two academics. The researcher first explained the DAG structure, the relationships between variables, and the measurement conditions used for each variable.

The FGD was conducted as one activity in which the 20 participants were divided into two groups to assess the probabilities for Cluster A and Cluster B. Tourism village managers were assigned to the group corresponding to the cluster of their tourism village, while the other stakeholders were distributed between the two groups. The Cluster A group consisted of one representative from the Tourism Office, one representative from Development Planning Agency, two Tourism awareness Group, four tourism village managers, and one academic. Meanwhile, the Cluster B group consisted of one representative from the Tourism Office, one representative from Development Planning Agency, two tourism awareness groups, six tourism village managers, and one academic. The tourism village managers and Tourism Office representatives involved in the FGD were the same participants who had been involved in confirming the DAG in the previous stage. Both groups used the same DAG structure, measurement conditions, questionnaire, probability assessment questions, and assessment procedure. The same DAG was used because both clusters represent tourism villages within the same Mount Merapi tourism system.

Although both groups assessed the same questions, the probability values were assessed separately based on the specific conditions of each cluster. The use of the same DAG, questionnaire, probability assessment questions, and assessment procedure was intended to maintain consistency between the two groups, although no additional

formal cross-group calibration was conducted.

The probability assessment was conducted collectively within each group. Participants directly provided percentage values ranging from 0% to 100% based on their knowledge, experience, and understanding of the tourism village conditions. If different values were proposed, the participants first discussed the differences. When an agreement was not directly reached, the value receiving the most votes was selected and then confirmed again by the group to obtain a collective agreement. All participants had equal weight in the assessment, regardless of their position or institutional background.

The probability values were directly obtained from the FGD participants rather than converted from qualitative statements by the researcher. The agreed percentage values were entered into the Bayesian Network model, where the GeNIe software system represented them on a probability scale from 0 to 1. For node without parent node, the FGD participants directly assigned probability of each state. Table 4 shows an example of the probability assessment for "internal policies" as a node without parent node for Cluster A.

Table 4. Prior probability for internal policies

Internal Policies	No	Yes
	0.6	0.4

For node with parent node, the FGD participants directly assessed the probability of each state of the child node for every combination of the parent states. The probability values of all child states had to total 100%. For nodes with more than two states, probability values were assigned to each state using the same procedure. The following example illustrates how participants assign probability values to child node states for each combination of parent node states during the FGD: if "economic activities" is in the "Limited" state, "agrotourism" is in the "Bad" state, "cultural preservation" is in "Passive", what is the probability that "resilience" is in the "Unstable or Stable" state? (Table 5).

Table 5. Conditional probability for resilience of tourism village

	Economic activities	**Limited**						**Adequate**					
Resilience of Tourism Village	Agrotourism	Bad		Moderate		Good		Bad		Moderate		Good	
	Cultural preservation	Passive	Active	Passive	Active	Passive	Active	Passive	Active	Passive	Active	Passive	Active
	Unstable	0.8	0.7	0.7	0.6	0.6	0.5	0.7	0.6	0.5	0.5	0.5	0.2
	Stable	0.2	0.3	0.3	0.4	0.4	0.5	0.3	0.4	0.5	0.5	0.5	0.8

Tables 4 and 5 show illustrative example of the probability assessment for node without parents and a child node with parents in Cluster A. The same procedure was repeated for every combination all nodes and parent state in both clusters. These probability values were then used to develop the conditional probability table (CPT) and calculate the initial probability conditions of the Bayesian Network model (Guo et al., 2026; Mun et al., 2025).

The consensus probability values were entered into the Bayesian Network, a strength analysis was conducted to assess the strength of the probabilistic relationships between variables (Ariyani & Fauzi, 2024). The strength of the relationship between parent and child nodes was assessed using the average Euclidean distance generated by the Strength of Influence function in GeNIe. Higher values indicate stronger relationship between the connected nodes. The analysis also reports the Maximum values as an additional summary of the relationship strength. Subsequently, a sensitivity analysis using a tornado chart was performed to identify the variables associated with the largest changes in the probability to the target node (Singto et al., 2020). The results were then used as a basis for developing probability scenarios through forward and backward propagation.

(3) Forward and backward propagation

The probability scenario was developed by setting the variables used as evidence with evidence of 100%. This step is carried out in two different directions. First, in forward propagation, two to three variables were set to 100% as evidence based on a joint decision and the results of the sensitivity analysis. Second, in backward propagation, the target variable represent the research objective was first set to 100% as evidence.

3. Results and Discussion

3.1 Analysis of Strength

This analysis is the relationship between variables in the Bayesian Network structure, where the analysis can be used to identify the strength of relationships between variables. Thicker arrows indicate a stronger relationship

between variables, such as parents and children, compactness's (Fauzi & Ariyani, 2024). The comparison of the probability values of the initial conditions and the strength of analysis obtained by the tourism villages of cluster A and cluster B is shown in Figures 5 and 6.

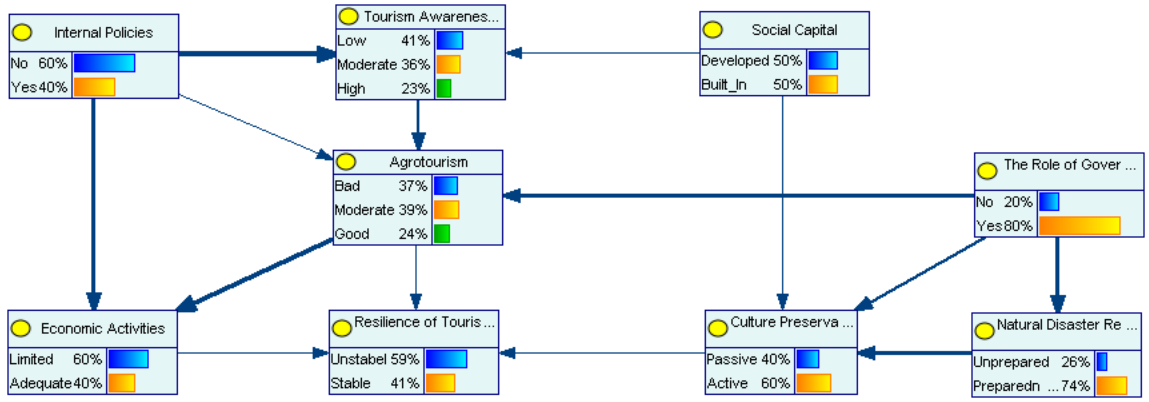


Figure 5. Initial probability of Bayesian Network for tourism village in Cluster A

The initial probabilities of the parent nodes in Cluster A show an 80% probability for the "Yes" state of the role of government (Figure 5). Social capital has an equal probability between the "Developed" and "Built-in" states at 50% each, while "Yes" state for internal policies, including AD/ART used by tourism villages in Cluster A, has a probability of only 40%. Under these initial conditions, the probability of the "Stable" state of resilience of tourism village is 41%. Therefore, the "Unstable" state has the higher initial probability in Cluster A.

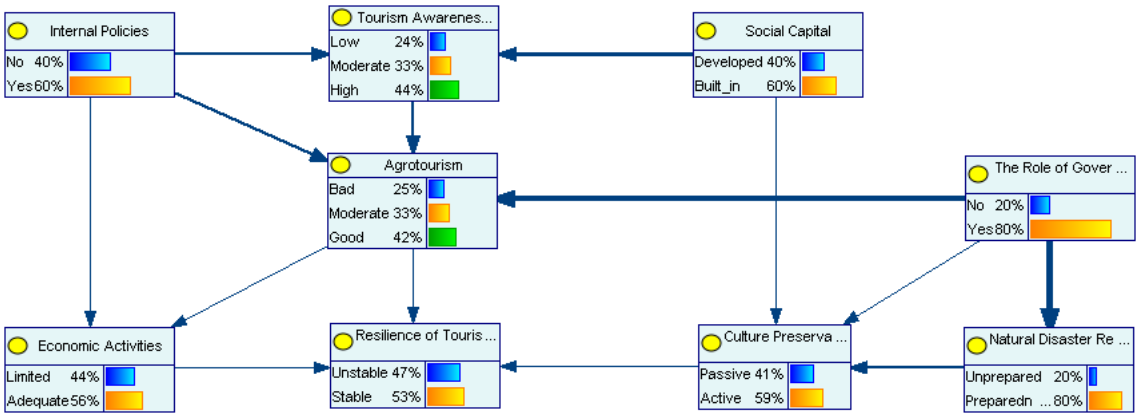


Figure 6. Initial probability of Bayesian Network for tourism village in Cluster B

In contrast to Cluster A, Figure 6 shows that the initial probability of the "Yes" state for the role of government in Cluster B is 80%. Social capital shows a 60% probability for the "Built-in" state, while the probability of the "Yes" state for internal policies is also 60%. Under these initial probability conditions, the probability of the "Stable" resilience of tourism village state in Cluster B is 53%. This indicates that Cluster B has a higher initial probability of the "Stable" resilience of tourism village state than Cluster A. The level of resilience of tourism villages in Clusters A and B is related to the relationship among the nodes or variable presented in Table 6.

In Cluster A, the role of government shows a relatively strong relationship with agrotourism, with an average strength value of 0.37 and a maximum of 0.46. It also has a relationship with community response to natural disasters response, with an average and maximum value of 0.30, and with cultural preservation, with an average value of 0.25 and a maximum of 0.30. Internal policies also related to economic activities, with an average value of 0.40 and a maximum of 0.70. The initial probability of the "No" state for internal policies reaches 60%, while economic activities are more likely to be in the "Limited" state. Internal policies are also related to tourism awareness group, with an average strength value of 0.42 and a maximum of 0.50. Tourism awareness group is also related to agrotourism, with an average value of 0.20 and a maximum of 0.36. Social capital in Cluster A does not show a relatively relationship strength with the other nodes. This condition is one of the differences between Cluster A and Cluster B.

Table 6. Strength analysis

Parents	Child	Cluster A		Cluster B	
		Average	Maximum	Average	Maximum
Internal policies	Tourism awareness group	0.42	0.50	0.23	0.27
Internal policies	Economic activities	0.40	0.70	0.20	0.30
The role of government	Agrotourism	0.37	0.46	0.42	0.60
Natural disaster response	Cultural preservation	0.35	0.50	0.25	0.40
Agrotourism	Economic activities	0.33	0.50	0.20	0.40
The role of government	Natural disaster response	0.30	0.30	0.50	0.50
The role of government	Cultural preservation	0.25	0.30	0.10	0.20
Tourism awareness group	Agrotourism	0.20	0.36	0.23	0.37
Agrotourism	Resilience	0.17	0.40	0.13	0.33
Internal policies	Agrotourism	0.17	0.36	0.21	0.36
Economic activities	Resilience	0.15	0.30	0.15	0.30
Social capital	Cultural preservation	0.15	0.30	0.20	0.30
Social capital	Tourism awareness group	0.13	0.27	0.32	0.35
Cultural preservation	Resilience	0.12	0.30	0.18	0.30

In Cluster B, the role of government also shows a strong relationship with agrotourism, with an average strength value of 0.42 and a maximum of 0.60. The relationship with community response to natural disasters response has an average and maximum value of 0.50. Unlike Cluster A, social capital in Cluster B shows a higher relationship strength with the tourism awareness group, with an average value of 0.32 and a maximum of 0.35. Internal policies are also related to the tourism awareness group, with an average value of 0.23 and a maximum of 0.27, and to agrotourism, with an average value of 0.21 and a maximum of 0.36.

3.2 Analysis of Sensitivity

Sensitivity analysis was used to examine how changes in related variables are associated with changes in the probability of the resilience target node. The green color bar shown in the tornado diagram (Figure 7 and Figure 8) represents a positive change, while the red bar represents a negative change (Fauzi & Ariyani, 2024).

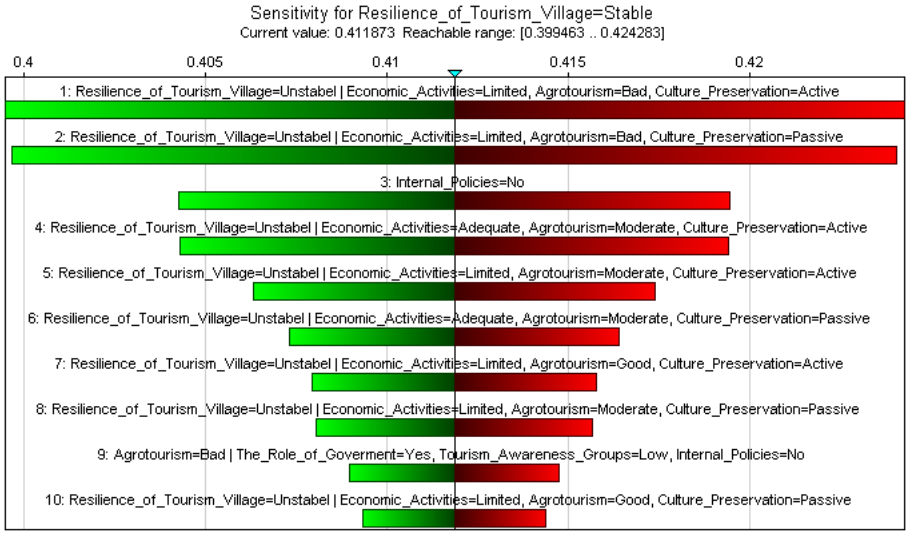


Figure 7. Tornado diagram of Cluster A tourism village

Based on Figure 7, the probability of the "Stable" state of resilience of tourism village in Cluster A is sensitive to changes in economic activities, agrotourism, and cultural preservation. A lower probability of the "Stable" state is associated with the "Limited" state of economic activities and the "Bad" state of agrotourism. Cultural preservation also shows different probability conditions, with both the "Active" and "Passive" states associated with a higher probability of the "Unstable" state under different conditions. These results indicate that changes in the states of economic activities, agrotourism, and cultural preservation are associated with notable changes in the probability of the "Stable" state in Cluster A.

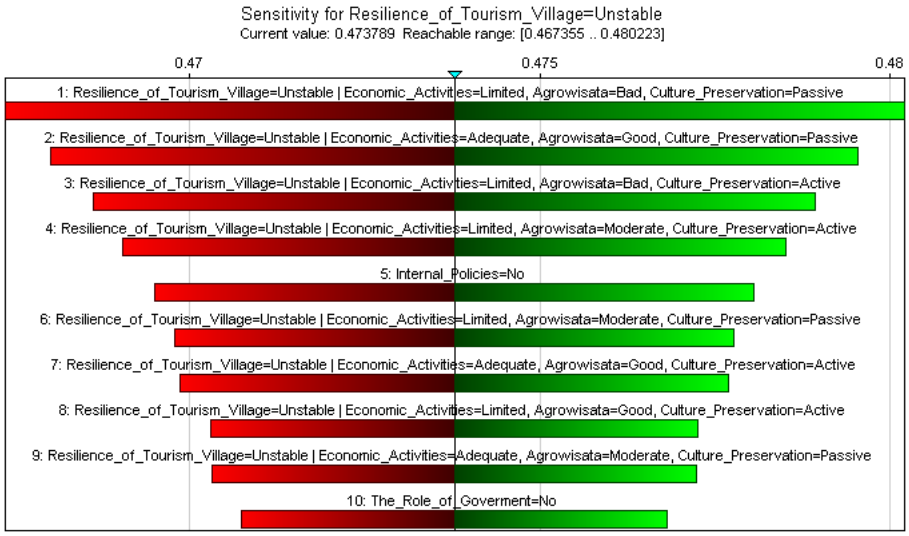


Figure 8. Tornado diagram of Cluster B tourism village

Based on Figure 8, the probability of the "Stable" state of resilience of tourism village in Cluster B is also sensitive to changes in economic activities, agrotourism, and cultural preservation. The first stem shows that a lower probability of the "Stable" state is associated with the "Limited" state of economic activities, the "Bad" state of agrotourism, and the "Passive" state of cultural preservation. These results show that the same three variables are sensitive in both clusters, although the probability conditions of their states different.

3.3 Probabilistic Scenario

The probabilistic scenario analysis was then conducted by setting the selected node states at 100% as evidence in GeNIe (Figures 9 and 10).

3.3.1 Probabilistic scenario with forward propagation

The first scenario was conducted using forward propagation by setting several selected node states as evidence based on the results of sensitivity analysis.

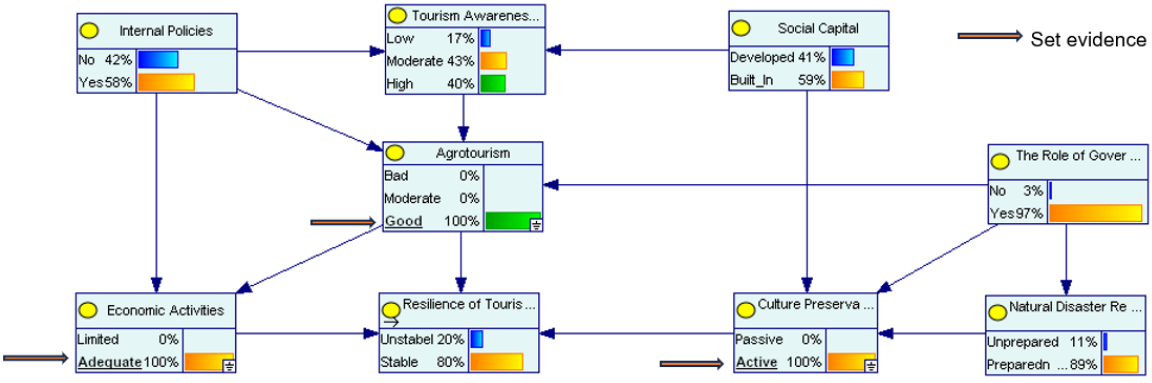


Figure 9. Cluster A of forward propagation

By setting the nodes for "Adequate" economic activities, "Good" agrotourism, and "Active" cultural preservation at 100% as evidence, the posterior probability of the "Stable" resilience of tourism village status was 80%, compared with 41% in the initial condition for Cluster A.

The same approach was also applied to Cluster B. By setting the nodes of "Adequate" economic activities, "Good" agrotourism, and "Active" cultural preservation at 100% as evidence, the posterior probability of the "Stable" resilience of tourism village status was 80%, compared with 53% in the initial condition.

A set of evidence was applied using the selected variables identified from the sensitivity analysis. These variables were selected because they showed stronger relationships with changes in the probability of the target node. This approach is supported by Gómez-Villegas et al. (2014) who used informative variables as evidence

variables to reduce uncertainty in the target.

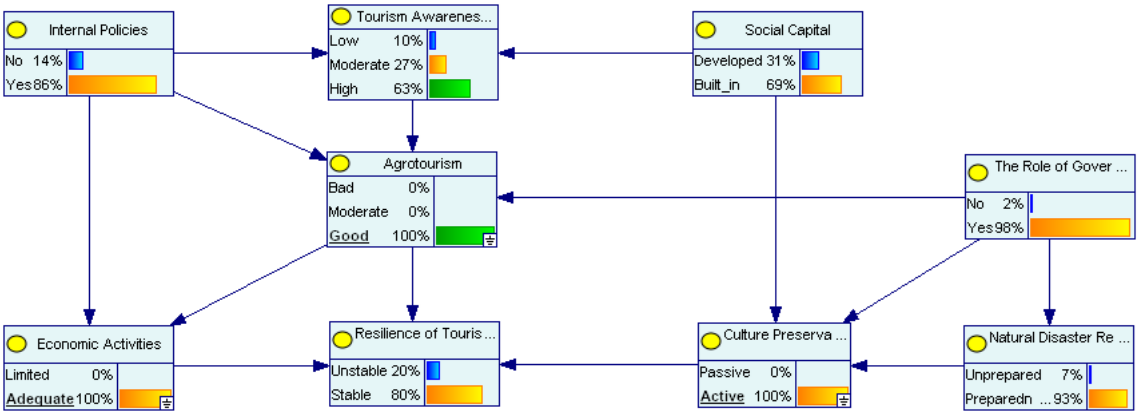


Figure 10. Cluster B of forward propagation

3.3.2 Probabilistic scenario with backward propagation

Backward propagation was conducted by setting the resilience target node in the desired "Stable" state at 100% as evidence in GeNIe. This analysis was used to identify the probability conditions associated with the desired target state.

Resilience of tourism village target node set to the "Stable" state at 100% as evidence. Figure 11 shows the probability conditions associated with this desired target state. These conditions include a "Moderate" tourism awareness group, "Built-in" social capital, "Moderate" agrotourism, "Yes" role of government, "Preparedness" natural disaster response, "Active" cultural preservation, and "Adequate" economic activities.

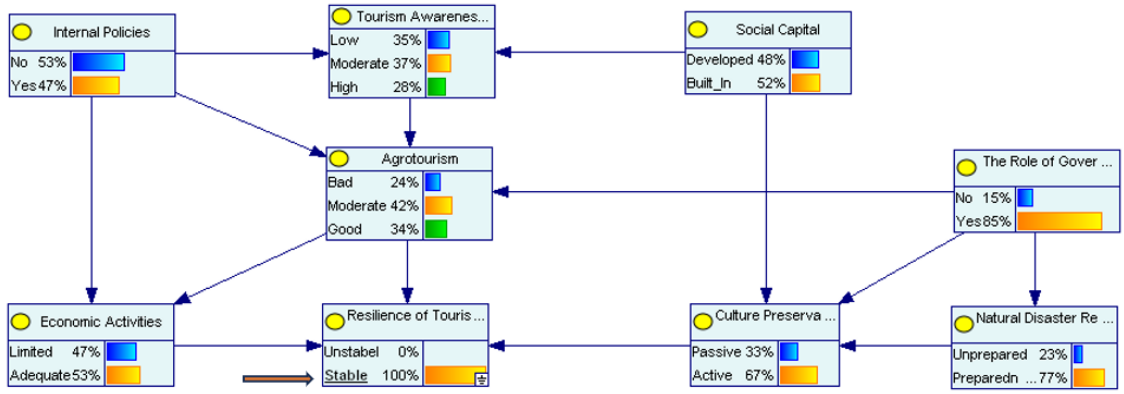


Figure 11. Cluster A of backward propagation

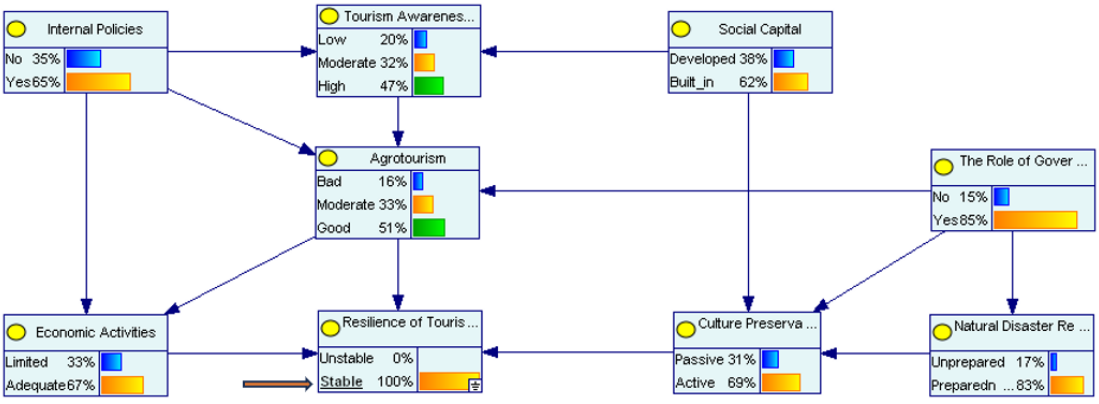


Figure 12. Cluster B of backward propagation

In the same way, by setting resilience of tourism village target node in Cluster B to the "Stable" state at 100% as evidence. Figure 12 shows the probability conditions associated with this desired target state. These conditions include "Yes" internal policies, a "High" tourism awareness group, "Built-in" social capital, "Good" agrotourism, "Yes" role of government, "Preparedness" natural disaster response, "Active" cultural preservation, and "Adequate" economic activities.

The comparison between Cluster A and Cluster B shows different probability conditions associated with the same desired "Stable" resilience of tourism village target state. Changes in the probability values can also be observed between the initial condition, forward propagation, and backward propagation. These differences are presented in Table 7.

Table 7. Comparing probabilistic scenarios

Variable	Measurement	Cluster A			Cluster B		
		Prior Probability	Forward Propagation	Backward Propagation	Prior Probability	Forward Propagation	Backward Propagation
Economic activities	Limited	60%	0%	47%	44%	0%	33%
	Adequate	40%	100%	53%	56%	100%	67%
Agrotourism	Bad	37%	0%	24%	25%	0%	16%
	Moderate	39%	0%	42%	33%	0%	33%
	Good	24%	100%	34%	42%	100%	51%
Cultural preservation	Passive	40%	0%	33%	41%	0%	31%
	Active	60%	100%	67%	59%	100%	69%
Social capital	Developed	50%	41%	48%	40%	31%	38%
	Built-in	50%	59%	52%	60%	69%	62%
The role of government	No	20%	3%	15%	20%	2%	15%
	Yes	80%	97%	85%	80%	98%	85%
Natural disaster response	Unprepared	26%	11%	23%	20%	7%	17%
	Preparedness	74%	89%	77%	80%	93%	83%
Internal policies	No	60%	42%	53%	40%	14%	35%
	Yes	40%	58%	47%	60%	86%	65%
Tourism awareness group	Low	41%	17%	35%	24%	10%	20%
	Moderate	36%	43%	37%	33%	27%	32%
	High	23%	40%	28%	44%	63%	47%
Resilience of tourism villages	Unstable	59%	20%	0%	47%	20%	0%
	Stable	41%	80%	100%	53%	80%	100%

The comparison of probabilistic scenarios in Table 7 shows different patterns between Cluster A and Cluster B. In the prior condition, Cluster B has a higher probability of resilience stability than Cluster A. After forward propagation, both clusters reach the same resilience stability probability of 80% when "Adequate" economic activities, "Good" agrotourism, and "Active" cultural preservation are set to 100%. However, the probability conditions of the other variables remain different between the two clusters. In backward propagation, resilience of tourism village target is set to the "Stable" state at 100% for both clusters, but each cluster shows different probability conditions among the related variables. These results indicate that the two clusters can move toward the same resilience target through different probability structures.

3.4 Discussion

The results of the Bayesian Network analysis were presented in the previous section. This study employed the model for exploratory and scenario-based purposes. Several key findings warrant further discussion to clarify differences between the two clusters and the factors associated with tourism village resilience. Accordingly, the probability outputs should be interpreted as conditional states represented within the model rather than as forecasts of future conditions. A limitation of this study is that no external or predictive validation was conducted. The probability estimates were derived primarily from stakeholder judgments gathered through the FGDs. Thus, the difference between Cluster A and Cluster B may partly reflect differences in stakeholder assessments between the two groups, rather than solely differences in the actual conditions of the clusters themselves.

Initial probability estimates indicate that Cluster B is more likely to begin in a stable resilience state than Cluster A. This gap is primarily associated with the parent-node conditions of internal policies and social capital in the Bayesian Network. In Cluster A, internal policies are largely draft rules and tend to be applied mainly when tourism activities or visits occur. Social capital is also split evenly between the "Built-in" and "Developed" states, suggesting that trust and cooperation exist but that participation often still requires encouragement, motivation, or incentives to remain active. By contrast, tourism villages in Cluster B already use internal policies as operational

guidelines for managing tourism, including task allocation, responsibilities, and profit sharing. Their social capital is more strongly "Built-in", meaning trust and cooperation are embedded in routine community activities and mainly need to be sustained. Prior studies likewise emphasize the importance of trust, cooperation, and community networks for community resilience and community-based tourism (Martinus et al., 2024; Musavengane & Kloppers, 2020). Government-related conditions show broadly similar probabilities across the two clusters. Accordingly, Cluster B's higher initial resilience probability appears to be driven mainly by more established internal policies and more embedded social capital.

This study's use of "Stable" and "Unstable" states simplifies the inherently dynamic nature of tourism village resilience into a probabilistic target. Therefore, the probability of the "Stable" state should be interpreted in relation to the variables and dependencies represented in the Bayesian Network, rather than as a comprehensive measure of resilience. Finally, applying the same DAG structure to both clusters assumes that variable relationships are similar across clusters and that differences are captured only through probability distributions—an assumption that may obscure cluster-specific relationships.

The high sensitivity of economic activities, agrotourism, and cultural preservation in both clusters can be understood through their close links to local resources and the core operations of tourism villages. Economic activities matter because tourism generates direct benefits for local residents through accommodation, food services, MSMEs, local products, tourism services, and employment. Prior studies similarly note that rural tourism can diversify livelihoods and expand local economic opportunities (Su et al., 2022). In the study area, agriculture and plantations remain key resources—particularly salak plantations in several villages—making agrotourism important because these assets can be transformed into tourism products and visitor experiences. At the same time, farming is seasonal, which can disrupt both tourism continuity and local livelihoods (Su et al., 2022). In the Bayesian Network, agrotourism relates to resilience both directly and indirectly via economic activities, indicating its dual role as an attraction and as a driver of income derived from agricultural resources. Cultural preservation is also sensitive because traditions, arts, and cultural practices are core endowments that shape village identity and distinctiveness. This makes the maintenance and strengthening of local cultural values important in both clusters, so they are not gradually eroded as tourism changes. Previous work likewise highlights sociocultural resources and heritage identity as contributors to rural community resilience (Gocer et al., 2024). However, cultural assets must be managed carefully, since excessive tourism use can undermine cultural values and social balance (Makwindi & Ndlovu, 2022). Overall, these three variables are sensitive because they underpin economic benefits, destination attractiveness, and the continuity of tourism village activities.

Sensitivity results also suggest that economic activities, agrotourism, and cultural preservation do not operate in isolation; they interact with social capital and government support. These linkages are particularly critical in disaster-prone tourism villages, where both community capacity and institutional assistance are needed before and after hazardous events.

Government roles and social capital play complementary roles across both clusters, especially under ex-ante and ex-post disaster conditions. Ex ante, government involvement extends beyond tourism development to include infrastructure, mitigation technologies, early-warning systems, information, training, and evacuation planning. In several villages around Mount Merapi, warning alarms are crucial because some tourism activities occur near river channels exposed to volcanic hazards; once warnings are issued or alarms are triggered, activities can be suspended until conditions improve. Effective communication between government agencies and communities strengthens this response: authorities provide training and evacuation-route information, while residents participate, share knowledge, and rotate attendance—reflecting social capital in the diffusion of information and the maintenance of preparedness. Thus, in ex-ante settings, government contributes infrastructure, technology, and technical knowledge, while social capital enables their uptake through trust, cooperation, and sustained participation.

In the post-eruption phase, social capital plays a key role in recovery. Residents can collaborate to clean homes and surrounding areas, provide temporary shelter for affected households, and restart community activities. Tourism villages with stronger finances may also draw on reserve funds to resume tourism management and support local residents. At the same time, government assistance remains essential for restoring infrastructure and livelihoods. This indicates that the relative importance of resilience factors shifts with volcanic conditions: government support and disaster response tend to be more critical for preparedness, while social capital and economic activity become especially important during recovery. This study does not explicitly model the temporal dynamics of Mount Merapi's eruption cycle (e.g., phases or alert levels) because the Bayesian Network is designed to capture the social, economic, institutional, cultural, and disaster-response conditions of tourism villages.

Overall, the findings suggest that tourism-village resilience depends on both community capacity and government support—an especially important insight in a disaster-prone destination such as the Mount Merapi region. Prior research on rural and community-based tourism has largely emphasized sustainability through community participation, local economic benefits, cultural and environmental preservation, and local governance. These elements also emerge here. However, in Merapi's tourism villages, sustainability efforts are insufficient without the capability to prepare for, respond to, and recover from volcanic disruptions. Accordingly, this study places greater emphasis on resilience as a foundation for tourism sustainability. This aligns with earlier Merapi

studies highlighting the importance of community adaptation and cooperation, disaster preparedness, and institutional support for recovery and for maintaining tourism after eruptions. In short, while sustainability concerns long-term tourism development, resilience is indispensable in hazard-prone destinations because villages must also be able to continue operating or rebound when shocks occur.

Although the two clusters show different probability patterns, several shared priorities stand out for strengthening tourism-village resilience. The findings offer practical implications for tourism-village management, community capacity building, and disaster preparedness in the Mount Merapi area and other high-risk tourism regions.

Key factors—economic activities, agrotourism, cultural preservation, social capital, internal policies, government roles, and natural disaster response—should be addressed in an integrated way. At the village level, internal policies can be reinforced through clear organizational structures, defined task divisions, transparent profit-sharing arrangements, and coordinated tourism services (e.g., a one-stop management system). These measures can strengthen local economic activity and broaden equitable participation in tourism.

For Mount Merapi and similar volcanic destinations, policy interventions should reflect the realities of tourism in hazard-prone landscapes. While volcanic features and local resources are valuable attractions, tourism remains exposed to ongoing risk. Tourism development should therefore be accompanied by a comprehensive disaster plan integrated into tourism-village governance. This plan should include early warning systems, disaster information and communication protocols, evacuation routes, mitigation training, procedures for temporarily suspending tourism during heightened risk, and post-disaster recovery arrangements. Effective implementation requires coordination among tourism village managers, local communities, tourism awareness groups, government agencies, and disaster management institutions, since preparedness and risk communication depend on strong stakeholder engagement. In this way, policy can better balance tourism growth with the capacity to prepare for, respond to, and recover from future disasters.

Finally, interventions can be structured as short- and long-term actions. In the short term, village managers, local governments, and disaster agencies can prioritize warning dissemination, evacuation planning, mitigation training, communication procedures, and clear protocols for temporarily halting tourism when risk escalates. In the long term, village managers, communities, tourism groups, and government institutions should strengthen internal governance, social capital, local economic activities, cultural preservation, and institutional coordination to sustain resilience and support long-term tourism sustainability.

4. Conclusions

Tourism villages in the Mount Merapi area have faced external shocks, disruption, and recovery periods while continuing to operate and support local communities. However, because they are located in a disaster-prone region, long-term sustainability will continue to depend on strong resilience. This study finds that although Clusters A and B differ in their probability conditions, both share several priorities: strengthening economic activities, agrotourism, cultural preservation, social capital, internal policies, government roles, and natural disaster response. Effective relationships between local communities and government institutions are also crucial for preparedness and for enabling recovery when disturbances occur. Strengthening these conditions in parallel is therefore essential to enhance tourism village resilience and support the long-term sustainability of tourism villages in the Mount Merapi area.

Author Contributions

Conceptualization, A.H.J., A.F., E.R., and W.R.; methodology, A.H.J. and A.F.; software, A.H.J. and A.F.; formal analysis, A.H.J., A.F., E.R., and W.R.; data curation, A.H.J., A.F., E.R., and W.R.; original draft preparation, A.H.J., A.F., E.R., and W.R.; review and editing, A.H.J., A.F., E.R., and W.R.; supervision, A.H.J., A.F., E.R., and W.R.; project administration, A.H.J.; funding acquisition, A.H.J. All authors have read and agreed to the published version of the manuscript.

Funding

This work is funded by Universitas Sebelas Maret, Surakarta, Indonesia through PDD-UNS scheme (Grant No.: 369/UN27.22/PT.01.03/2025).

Ethical Approval

The study was approved by Health Research Ethics Committee, Ministry of Health, Semarang Health Polytechnic (approval number: 1318/EA/F.XXIII.38/2025; date: December 19, 2025).

Data Availability

The processed clustering results are presented in the manuscript. Examples of the prior probability assessment and conditional probability table (CPT) are also provided to illustrate the probability elicitation procedure. The complete clustering input data, stakeholder probability assessments, full CPTs, and Bayesian Network model are not publicly available due to confidentiality considerations. Access to these data is restricted and may only be considered upon reasonable request, subject to confidentiality requirements and approval by the research team.

Acknowledgments

The authors are grateful to the Government of Sleman Regency and the managers of local tourism villages for their cooperation and willingness to participate in this study.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors used ChatGPT solely for minor language editing to improve grammar, clarity, and readability. All scientific content, data analysis, interpretations, and conclusions were developed by the authors.

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